

IMAGE PROCESSING TECHNIQUES TO IMPROVE APRIL TAG POSE ESTIMATION IN LOW-COST SYSTEMS

April Tag Estimation - Connor Guzikowski, Carson Rae, Derrick Eliasi

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ABSTRACT

Accurate and efficient pose estimation of fiducial markers is essential for robust localization and mapping in robotic systems. This project aims to improve the accuracy and computational performance of AprilTag pose estimation in both single-image use cases. In the first phase, we employ an open source AprilTag pose estimation library developed to perform controlled single image tests under varying conditions of viewing angle, distance, fog, lighting, and motion blur. In these tests, we perform various image processing techniques such as gamma correction, histogram equalization, Retinex-based illumination correction, Gaussian and bilateral filtering for denoising, and deblurring through Wiener and blind deconvolution methods. Additionally, we apply contrast-limited adaptive histogram equalization (CLAHE) and color balance adjustments. These experiments will quantify how such different image processing algorithms influence both pose accuracy and processing speed. To evaluate performance changes with image processing techniques, we will measure improvements in detection rates compared to the baseline system. By jointly analyzing confidence and speed trade-offs across single-image contexts, this work aims to find improvements to an already useful system without sacrificing processing speed.

Index Terms— AprilTag, fiducial markers, pose estimation, image processing, low light, fog, motion blur, robotics

1. INTRODUCTION

AprilTags are one of the most common tools used for pose estimation in robotics because they provide a fast and reliable way to calculate a camera’s position and orientation. They work well in controlled environments, but in real-world conditions the accuracy can drop significantly when lighting or visibility changes. Low-light scenes, fog, glare, and motion blur all reduce the contrast between the black and white tag regions, making it harder for detectors to locate corners accurately. These small detection errors can accumulate in systems like TagSLAM [1], where pose estimation depends on consistent tag observations over multiple frames.

The TagSLAM research combines tag detections across many frames using a factor graph optimization approach in

order to extend the AprilTag framework. It provides accurate pose estimation in normal imaging conditions, but it assumes that the tag detections are always reliable. Moreover, low-cost cameras used in robotics, such as Raspberry Pi cameras, often operate and struggle under poor lighting or environmental interference. In these settings, TagSLAM’s accuracy is negatively affected. These problems are even more noticeable in low-power systems that do not have extra sensors or built-in lighting to help with detection.

To address these challenges, our project aims to improve April Tag detection and pose estimation by integrating image processing before tag detection. The focus is to evaluate how image enhancement methods affect single-frame accuracy. We plan to test the system under three main visual conditions that typically reduce performance: low-light and harsh shadow environments, motion blur, and foggy or hazy scenes. Each of these conditions introduces unique image distortions that interfere with accurate tag detection and reliable pose estimation.

In this project, we will test and compare different image enhancement methods, such as histogram equalization, Retinex-based correction, dehazing, and contrast normalization, to see which help the most to improve tag visibility and stability. After enhancing the images, we will add these versions to the April Tag pipeline and check how they affect pose accuracy, detection, and runtime. Ultimately, our goal is to make tag detection more reliable in real-world conditions while keeping the system fast and lightweight for low-cost robots.

2. RELATED WORKS

2.1. AprilTag Estimation

Several prior studies have sought to improve aspects of the April Tag based pose estimation, including accuracy, robustness, and computational latency. One of particular interest is the work done by Primmer and Daniilidis [1], who introduced *TagSLAM*, a system that integrates fiducial markers into a full Simultaneous Localization and Mapping framework. Unlike normal AprilTag detection pipelines that operate on individual frames, TagSLAM creates a factor graph that merges observations from multiple cameras and markers over time. This formula allows for global optimization

of both camera and tag poses, significantly decreasing drift and performing with greater spatial consistency across larger environments. Their results demonstrated that incorporating fiducial markers in a graph-based optimization pipeline leads to more reliable localization, specifically in indoor and spaces without GPS.

While TagSLAM focuses on high-level pose graph optimization and mapping accuracy, its performance builds heavily upon the quality of the raw tag detections feeding into it's system. Previous works, such as Olson's foundational April-Tag system [2], emphasize detection speed and tag design for geometric strength, but they typically assume ideal image conditions. As a result, both frameworks are at-risk to degradation in quality under challenging visual scenarios such as fog, motion-blur, or low light. All of which, are conditions frequently encountered by UAV and robotic systems. This project builds upon the TagSLAM framework by addressing these limitations at the image-processing level, introducing preprocessing methods aimed at increasing the reliability of AprilTag detections prior to pose estimation.

2.2. Condition 1: Foggy Conditions

2.2.1. Enhancement Based Defogging Techniques

Enhancement-based defogging approaches aim to improve image visibility in foggy conditions through classical enhancement operations such as contrast stretching, histogram equalization, Retinex filtering, multi-scale fusion, and edge-preserving filtering. These methods focus on increasing local contrast and restoring suppressed color and texture features rather than modeling the physical scattering process that causes fog [3].

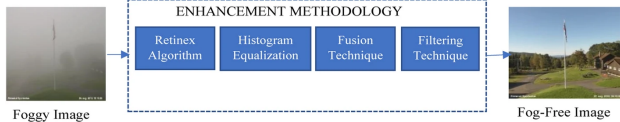


Fig. 1: Enhancement Based Image Processing Pipeline [3]

While these methods are computationally efficient and simple to implement, they often over-enhance or distort natural image statistics due to the absence of physical modeling. Consequently, enhancement-based techniques may produce oversaturated or geometrically inconsistent results, which can negatively impact downstream vision tasks such as corner detection and pose estimation in fiducial-based systems.

2.2.2. Prior-Based Defogging Techniques

Prior-based methods rely on the atmospheric scattering model and incorporate statistical or structural assumptions—known as priors—to estimate the transmission map and atmospheric light parameters [3]. These techniques balance physical inter-

pretability and computational practicality by enforcing constraints derived from empirical observations of natural images.

Common priors include the *Dark Channel Prior (DCP)* [4], which assumes at least one color channel has low intensity in fog-free patches, and the *Color Attenuation Prior (CAP)* [5], which models the relationship between scene depth and color differences in brightness and saturation. The *Haze Line Prior (HLP)* [6] captures pixel distributions along haze lines in RGB space, while the *Color Ellipsoid Prior (CEP)* and *Difference Structure Preservation Prior* [3] extend these concepts to better model pixel distributions and maintain structural consistency.

These physically grounded techniques typically produce more accurate and artifact-free results than enhancement-based methods, especially in scenes that follow the assumed priors. However, they may fail under conditions such as non-uniform fog, bright sky regions, or non-Lambertian surfaces, leading to halo effects or incorrect depth estimations. For real-time applications like pose estimation and SLAM, prior-based methods offer improved geometric fidelity but at a higher computational cost.

2.3. Learning Based Techniques

Deep learning approaches have recently become prominent in image defogging due to their ability to model complex nonlinear relationships between foggy and clear images [3]. These models use convolutional neural networks (CNNs) or generative adversarial networks (GANs) trained on large datasets to directly predict clear images or intermediate representations such as transmission maps.

Notable examples include *DehazeNet* [7], which estimates transmission maps in an end-to-end manner, and *Ranking-CNN* [8], which introduces ranking layers to preserve structural relationships between foggy and clear features. Multi-resolution and cascaded CNN architectures combine global and local features to improve robustness, while models like *FAMED-Net* [9] emphasize lightweight, real-time inference.

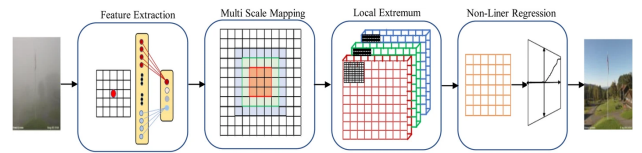


Fig. 2: DehazeNet Block Diagram [7]

Learning-based methods achieve high visual quality and generalization across fog densities but depend heavily on large, representative datasets. They are also prone to domain shift when trained on synthetic data and may introduce latency that limits real-time deployment. In fiducial-based vision systems, these techniques can significantly enhance tag visibility, though their effect on geometric accuracy must

be carefully validated.

2.4. Condition 2: Motion Blur

2.4.1. Improved Handling of Motion Blur in Online Object Detection

Sayed *et al.* [10] investigated the effect of motion blur on object-detection accuracy and proposed an adaptive framework that works to condition detection models on the specific type and degree of motion blur. Instead of attempting to simply clear all blur, their method incorporates blur information into the detection process itself. The authors worked to develop synthetic motion-blur augmentation and blur-conscious label generation to train convolutional networks to be capable of distinguishing blur patterns and compensating for them during inference. Experiments ran on the COCO dataset found that blur-aware models achieve a higher mean Average Precision (mAP) than standard detectors when tested on real blurred imagery, proving that blur-specific training can successfully mitigate accuracy loss.

This study is valuable because it looks at motion blur as a learning problem rather than a correctional problem. However, the approach is quite costly as it relies heavily on synthetic blur models and large-scale training, which is impractical for lightweight and on-the-go systems. While effective for object detection, it does not impact the geometric consistency needed for pose estimation. In contrast, this paper's proposed work focuses on pre-processing to restore image sharpness before detection, ensuring continued compatibility with existing AprilTag pose estimation pipelines without retraining detection networks.

2.4.2. Ghost-DeblurGAN for Fiducial Marker Detection

Liu *et al.* [11] introduced Ghost-DeblurGAN, a deep generative model designed to correct motion blur in images containing fiducial markers such as ArUco or AprilTag. The authors constructed a specialized dataset, YorkTag, that includes synthetic and real blurred images of markers under differing camera and object motions. The network uses a residual generator in combination with a perceptual loss that preserves the high-frequency texture details critical for reliable marker decoding. Experimental results showed vast improvement in marker detection rates and pose-estimation accuracy after deblurring, performing better than classical methods like Wiener and Richardson-Lucy filtering in both sharpness and downstream detection metrics.

This work is particularly relevant because it directly looks at deblurring with fiducial markers as a performance metric, rather than overall image quality. Its use of GAN-based restoration indicates that even subtle motion blur can drastically reduce detection reliability. However, the model's computational complexity and dependence on a large training set limits its portability to mobile robots, such as a UAV. The

present project builds on this concept by looking to explore more lightweight image-processing techniques that can mitigate motion blur without retraining large networks, aiming for real-time improvement of AprilTag pose estimation in dynamic robot environments.

2.5. Condition 3: Low Light/Harsh Shadows

2.5.1. Low-Light Enhancement for Object Detection

One relevant study by Guo *et al.* [12] proposed a dynamic end-to-end framework for object detection in low-light environments. Instead of following a standard process of enhancing the environment and then detecting objects, their method combines both steps into a single network. This allows the system to automatically choose which enhancement path to apply based on the lighting conditions and noise level of each image. Their experiments on the ExDark dataset showed improvements in detection accuracy, especially for medium-sized objects, while keeping the overall computation time reasonable.

This approach works well because it adjusts the enhancement on a per-image basis instead of using one fixed model. That makes it more reliable in uneven or harsh shadow lighting scenes. However, the model is more complex and takes longer to train, making it less ideal for low-power or real-time systems. Since the design is focused on object detection and not geometric localization, it would also need modification before being applied to tasks like AprilTag pose estimation.

2.5.2. Low-Light Enhancement for License Plate Recognition

Another relevant study addressing visibility degradation in dark scenes is by Saputra *et al.* [13], who proposed a low-light image enhancement pipeline for license-plate recognition using URetinx-Net and TRBA. Their approach demonstrates a Retinex-based deep neural network to separate illumination and reflectance components, followed by a text-recognition network to improve character clarity under illumination levels below 20 lux.

The paper displays how the URetinx-Net processing method improves image contrast and readability compared to simple enhancement methods such as histogram equalization and CLAHE. Using this method, they were able to reach higher recognition accuracy while maintaining stable color balance, making it suitable for embedded or mobile systems. Furthermore, using this approach worked well in different light settings, illustrating how Retinex-based deep networks can adapt to diverse nighttime scenes.

Although the method performs well for improving text visibility, it is mainly designed for 2-D license plate recognition rather than pose estimation. The approach also requires heavy computation and supervised training data, making it less practical for real-time or low-cost robotic systems.

3. PROPOSED METHODS

3.1. AprilTag Estimation

The AprilTag system is the starting point for our work, as it provides a simple and reliable way to estimate a camera’s position and orientation from a known marker. Our setup uses the open-source AprilTag 3 library, which is also used in the TagSLAM framework. TagSLAM builds on the normal AprilTag detector by combining tag observations from many frames using a factor graph, which helps reduce drift and improve overall consistency during motion.

In the basic AprilTag detection pipeline, the input image is first converted to grayscale and then thresholded to separate the black and white tag regions. Contours are found and grouped into quadrilateral shapes that match the tag layout. Each possible tag is then decoded using its unique binary ID pattern, which includes built-in error correction to avoid false detections even when light or noise affects the image.

Once the corners are detected, the 3D pose of the tag is calculated using a Perspective-n-Point (PnP) solver. With the known tag size and the intrinsic parameters of the camera, the system computes the transformation between the tag and the camera as:

$$\mathbf{T}_{cw} = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}, \quad (1)$$

where $\mathbf{R}$ represents the rotation and $\mathbf{t}$ represents the translation of the camera relative to the tag.

TagSLAM takes these individual AprilTag poses and refines them over multiple frames by optimizing all tag and camera positions together. This allows for smoother and more accurate trajectories, especially when the same tags are seen repeatedly. In our project, we use this setup as the baseline and then add different image enhancement techniques before the tag detection step. This lets us test how improved image quality can help TagSLAM detect corners more accurately and produce better overall pose estimates.

3.2. Condition 1: Defogging

In this project we will explore the impact of three defogging strategies: enhancement-based, prior-based, and learning-based techniques for AprilTag pose estimation accuracy and processing efficiency. Our goal is to evaluate how each approach affects fiducial detection reliability and runtime in both standalone and SLAM-integrated image sequences using TagSLAM.

3.2.1. Enhancement-Based Defogging via Retinex

For the enhancement based pipeline we will use a Python implementation of Single Scale Retinex (SSR) and Multi Scale Retinex (MSR) from the open-source project “Retinex-Image-Enhancement”. [14] Retinex theory is based on decomposing the observed image into illumination and re-

flectance components:

$$I(x) = R(x) L(x), \quad (2)$$

where $I(x)$ is the observed intensity at pixel x , $R(x)$ is the reflectance (scene content) and $L(x)$ is the illumination. In MSR, the enhanced reflectance estimate can be expressed as

$$R_{\text{MSR}}(x) = \sum_{n=1}^N w_n \left[\log I(x) - \log (I(x) * F_n(x)) \right], \quad (3)$$

where $F_n(x)$ are Gaussian kernels of varying scales, and w_n are weights summing to one. We will apply this implementation to foggy images of fiducial tags to boost contrast, reveal hidden edges and corners, and then apply our tag-detection pipeline. We will test a variety of variances to see what optimizes for speed and clarity. The key metrics will be pose error and frame processing time.

3.2.2. Prior-Based Defogging: Dark Channel Prior (DCP)

We will use an implemented Dark Channel Prior method from [15] following the classical atmospheric scattering model:

$$I(x) = J(x) t(x) + A (1 - t(x)), \quad (4)$$

where $I(x)$ is the observed foggy image, $J(x)$ is the scene radiance, A is global atmospheric light, and $t(x)$ is the transmission map. The dark channel of a clear image is defined as:

$$J_{\text{dark}}(x) = \min_{y \in \Omega(x)} \left(\min_{c \in \{r, g, b\}} J^c(y) \right), \quad (5)$$

and the transmission estimate:

$$t(x) = 1 - \omega \min_{y \in \Omega(x)} \left(\min_c \frac{I^c(y)}{A^c} \right), \quad (6)$$

with $\Omega(x)$ the local patch around pixel x . The recovered radiance is then:

$$J(x) = \frac{I(x) - A}{\max(t(x), t_0)} + A, \quad (7)$$

where t_0 is a lower bound on transmission to avoid division artifacts. Guided filtering is added for edge-aware smoothing of $t(x)$. The output will serve as input to the tag detection and pose solver.

3.2.3. Learning-Based Defogging: DehazeNet Implementation

We adopt the Feature Fusion Attention Network (FFA-Net) [?] as a learning-based defogging method. Unlike prior learning approaches that explicitly estimate a transmission map and atmospheric parameters, FFA-Net performs end-to-end image restoration, directly mapping a single foggy

image to its haze-free counterpart using a deep convolutional neural network. FFA-Net is composed of multiple residual groups with feature fusion attention (FFA) modules that combine both channel-wise and pixel-wise attention mechanisms. These attention modules enable the network to adaptively emphasize informative features while suppressing haze-corrupted regions, allowing effective handling of spatially varying fog density. By fusing features across multiple depths of the network, FFA-Net captures both local texture details and global scene structure, which is critical for restoring contrast and visibility under heavy haze conditions. Rather than relying on the explicit atmospheric scattering model,

$$I(x) = J(x)t(x) + A(1t(x))$$

FFA-Net implicitly learns the inverse degradation process from data, avoiding errors introduced by inaccurate transmission or atmospheric light estimation. We use the pretrained FFA-Net model provided by the authors and integrate it directly into our image processing pipeline. We will compare the performance of DehazeNet with our enhancement-based (Retinex) and prior-based (DCP) pipelines across both single-image tests and SLAM experiments to evaluate the trade-offs between accuracy, robustness, and runtime between the three.

3.3. Condition 2: Motion Blur

This section focuses on restoring motion-blurred images prior to AprilTag pose estimation. Two classical image restoration techniques, Inverse Filtering and Wiener Filtering are implemented and evaluated for their ability to correct linear motion blur. Both methods work in the frequency domain using a mathematically modeled degradation function $H(u, v)$ that represents the spread function (PSF) of motion blur aligned with the direction of the camera movement. See the following image for an example of what this project aims to accomplish via these filters:

3.3.1. Inverse Filtering

Inverse filtering utilizes a direct approach to deconvolution by assuming that the degradation function $H(u, v)$ is well known and invertible.

$$G(u, v) = H(u, v)F(u, v) \quad (8)$$

where $F(u, v)$ is the original image, and the restored estimate is obtained by division:

$$\hat{F}(u, v) = \frac{G(u, v)}{H(u, v)} \quad (9)$$

To avoid division by zero where $H(u, v)$ is small, a small constant or a low-pass filter is introduced:

$$\hat{F}(u, v) = \frac{H^*(u, v)}{|H(u, v)|^2 + \epsilon} G(u, v), \quad (10)$$

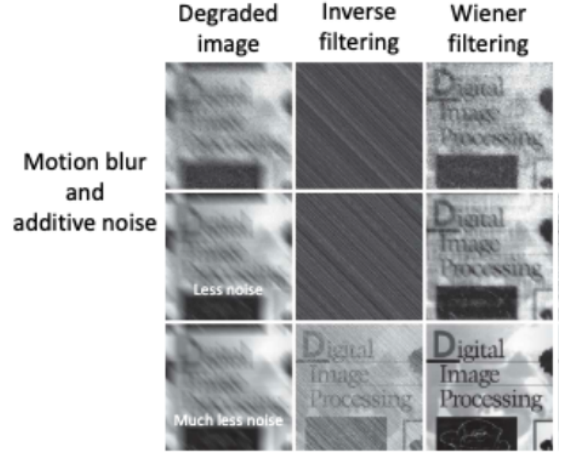


Fig. 3: Adapted from ECE 253 Lecture 7 Slides [16]

where ϵ is a stabilization term. The PSF $h(x, y)$ is constructed according to the motion length L and angle θ of the blur, and its Fourier Transform $H(u, v)$ is used for restoration. Inverse filtering can recover high-frequency detail effectively when noise is negligible, but it is highly sensitive to additive noise, often amplifying high frequency artifacts and ringing. Despite some limitations, this method works as a strong baseline for evaluating more robust approaches.

3.3.2. Wiener Filtering

The Wiener filter extends inverse filtering by incorporating a statistical model of noise and signal power, minimizing the mean-square error between the estimated and original images. It's frequency domain formula is given by:

$$\hat{F}(u, v) = \frac{H^*(u, v)}{|H(u, v)|^2 + \frac{S_n(u, v)}{S_f(u, v)}} G(u, v), \quad (11)$$

where $S_n(u, v)$ and $S_f(u, v)$ are the power of the noise and unblurred image, respectively. In practice, the ratio $\frac{S_n}{S_f}$ is approximated by a constant K , getting the simplified expression

$$\hat{F}(u, v) = \frac{H^*(u, v)}{|H(u, v)|^2 + K} G(u, v). \quad (12)$$

This formula suppresses noise amplification while maintaining edge correctness. When it is applied to motion blurred images, the Wiener filter achieves clearer and more visually correct reconstructions than inverse filtering. However, its effectiveness also depends on accurate estimation of the PSF and noise level. Overly large K values can lead to over smoothed results. In the context of AprilTag pose estimation, Wiener filtering can be expected to produce more stable tag detection under realistic noisy image conditions, providing a balance between sharpness recovery and noise suppression.

3.3.3. Pipeline Implementation

Following the theoretical groundwork above, we implemented a complete restoration-and-detection pipeline that operates on every motion-blurred entry in `dataset/data.csv`. The major stages are:

- **PSF synthesis & OTF generation:** For each frame we synthesize a linear motion PSF that matches the capture settings (15 px, horizontal) and convert it to its Fourier-domain degradation function $H(u, v)$. This PSF is reused for both filters so their differences stem purely from the reconstruction strategy.
- **PSF search caveats:** Early versions swept a grid of lengths/angles and chose the PSF with the highest Laplacian variance, but we observed two issues: (1) Laplacian variance tends to saturate at the ends of the grid (e.g., always preferring 15 px or 51 px) when the focus gain is small, and (2) frames without intentional blur were sometimes “sharpened” unnecessarily because the metric still nudged them toward a non-zero PSF. To keep the pipeline stable we eventually fixed the PSF to the known capture blur and added a fallback that reverts to the baseline whenever the Laplacian gain is negligible.
- **Multi-branch restoration:** We maintain three image branches—(i) the untouched baseline, (ii) a pseudo-inverse reconstruction using Eq. (5) with a stabilizing ϵ , and (iii) the Wiener reconstruction using Eq. (7) with $K = 10^{-3}$. All processing occurs in the frequency domain to mirror the formulations in the previous subsections.
- **Sharpness verification:** We compute the Laplacian variance of each restored frame and compare it against the baseline to ensure the PSF actually increases high-frequency content. If no improvement is observed we leave that frame in the “baseline” state so the pipeline never degrades a clean input.
- **AprilTag detection & comparison:** Each branch runs through the same detector (calibrated PiCam intrinsics, tag size 0.19 m). We record detection counts, decision margins, and pose depths for every branch, then label the frame with whichever branch produced the most confident detections (ties broken by decision margin). This “best-of” logic guarantees that overall tag counts cannot decrease.
- **Result logging:** For each frame we save a three-panel comparison figure, a JSON file containing the PSF parameters and per-branch detection stats, and append a row to a summary CSV. Companion scripts

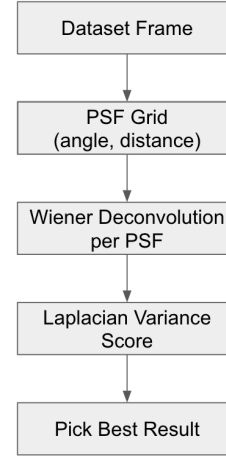


Fig. 4: Motion Blur Processing Pipeline

(`pose_table.py`, `plot_depth_table.py`) aggregate the JSON data into tables/plots showing ground-truth vs. estimated depths per algorithm despite gaps in detections.

This implementation turns classical inverse/Wiener formulas into a reproducible pipeline that documents when each method helps or hurts motion-blurred AprilTag pose estimation.

3.4. Condition 3: Low Light and Harsh Shadows

Low-light and shadow-heavy environments significantly degrade AprilTag detection performance. In such scenes, the contrast between tag edges becomes weak, corner localization becomes unstable, and TagSLAM’s downstream pose estimation accuracy suffers. To address this, we implemented and evaluated two different enhancement pipelines designed to restore contrast, recover shadow detail, and improve the robustness of AprilTag detection under challenging illumination.

3.4.1. Gamma, CLAHE, and Retinex Enhancement

Our first approach combined three classical enhancement techniques: gamma correction, CLAHE-based contrast stretching, and a simplified Retinex illumination-removal model. The goal of this pipeline was to lift shadowed regions while preserving natural appearance and avoiding over-amplification of noise.

We began by computing the mean and standard deviation of the grayscale image,

$$\mu_I = \frac{1}{N} \sum_{x,y} I(x, y), \quad \sigma_I = \sqrt{\frac{1}{N} \sum_{x,y} (I(x, y) - \mu_I)^2}, \quad (13)$$

and applied enhancement only when μ_I fell below a low-light threshold. Gamma correction with $\gamma \in [0.6, 0.8]$ brightened the darkest regions, after which CLAHE was applied to locally increase contrast and recover edges,

$$I'(x, y) = \text{HE}(I(x, y)^\gamma). \quad (14)$$

To further address uneven lighting, we incorporated a Retinex-style illumination normalization:

$$R(x, y) = \log(I(x, y)) - \log(G_\sigma * I(x, y)), \quad (15)$$

which isolates reflectance by removing low-frequency illumination. The final enhanced frame blended Retinex detail recovery with the gamma+CLAHE output,

$$I_{\text{enh}}(x, y) = \alpha R(x, y) + (1 - \alpha) I'(x, y), \quad (16)$$

using $\alpha \approx 0.7$.

3.4.2. Adaptive Shadow Boosting

While our gamma+CLAHE+Retinex pipeline improved visibility, its computational cost and occasional halo artifacts made it less suitable for real-time pose estimation. To address these shortcomings, we developed a custom *adaptive shadow boosting* method that selectively enhances only the underexposed regions of an image. This method is simple, fast, and highly effective for improving AprilTag detection in low-light or high-shadow environments.

Instead of applying global enhancement, we first identify which parts of the image are genuinely shadowed. Given the grayscale image $I(x, y)$, we construct a binary mask

$$S(x, y) = \begin{cases} 1, & \text{if } I(x, y) < \tau_{\text{shadow}}, \\ 0, & \text{otherwise,} \end{cases} \quad (17)$$

where τ_{shadow} is an empirically chosen threshold (set to 80 in our implementation). This allows the algorithm to focus enhancement where it is actually needed, preserving the original appearance of properly exposed regions.

Once the shadow mask is created, we apply a multiplicative boost only to shadowed pixels:

$$I_{\text{boost}}(x, y) = \begin{cases} \beta \cdot I(x, y), & \text{if } S(x, y) = 1, \\ I(x, y), & \text{otherwise,} \end{cases} \quad (18)$$

where β is our shadow amplification factor (set to 2.5). This targeted approach prevents the washed-out appearance that occurs when global brightness adjustments are applied uniformly.

To avoid sharp boundaries between boosted and non-boosted regions, we apply a bilateral filter, which smooths intensity while preserving edges:

$$I_{\text{smooth}} = \text{Bilateral}(I_{\text{boost}}). \quad (19)$$

Finally, we stretch the image contrast using the 2nd and 98th percentiles of pixel intensity:

$$I_{\text{final}}(x, y) = \text{clip} \left(\frac{I_{\text{smooth}}(x, y) - p_2}{p_{98} - p_2} \cdot 255, 0, 255 \right), \quad (20)$$

ensuring full utilization of the available dynamic range without amplifying noise.

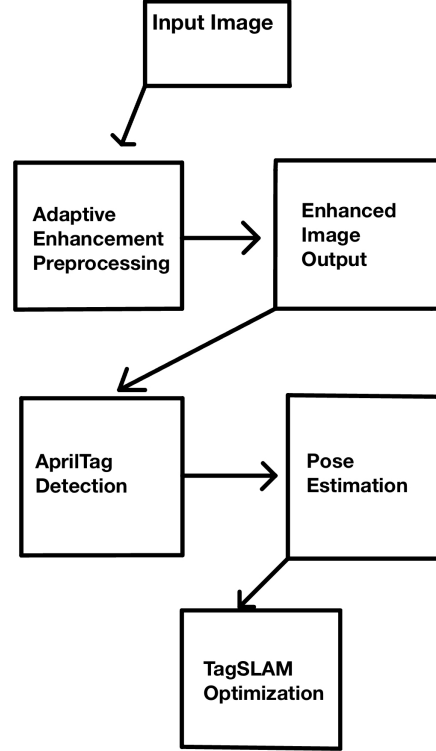


Fig. 5: Final preprocessing pipeline used to improve AprilTag detection and pose estimation under different conditions.

3.5. Integration into AprilTag Detection

After the image is enhanced, it will go directly into the AprilTag detector used. For each frame t , the detector gives the corner positions of the tag $\mathbf{z}_t$ and an estimated pose of the camera $\mathbf{T}_{cw,t} \in SE(3)$. The backend will then minimize the total reprojection error:

$$E = \sum_t \sum_i \|\pi(\mathbf{T}_{cw,t} \mathbf{P}_i) - \mathbf{z}_{i,t}\|^2, \quad (21)$$

where $\pi(\cdot)$ is the projection model and $\mathbf{P}_i$ are the tag corner points. By improving image contrast before detection, we saw smaller pixel errors and better pose accuracy during optimization. In addition to this, by automatically adapting the brightness and contrast of the image before detection runs, this method makes pose estimation more consistent in various

environments without slowing down the system. To measure improvements, we will compare the modified system with the original baseline in multiple test scenes using pose accuracy, detection rate, and computation time.

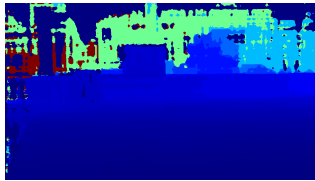
4. EVALUATIONS

4.1. Experiment Setup/Dataset

To evaluate our proposed enhancements to AprilTag pose estimation, we will test how they affect the speed and accuracy in single image tests of pose estimation. For these tests, we collected our own data, taking single images of AprilTags under varying conditions and from varying positions. We captured images in fog, under motion, and different lighting conditions to test the effectiveness of our methods. The images were captured with a low-cost Raspberry Pi camera. To get the ground truths of the poses of the images, we used an Oak-D Lite depth camera to measure how far away the april tags were. The Oak-D Lite and Raspberry Pi camera are fixed in position relative to each other using a 3D-printed jig.



(a) Oak D Lite RGB



(b) Oak D Lite Depth

Fig. 6: Images from Oak D Lite Camera



Fig. 7: Raspberry Pi camera image

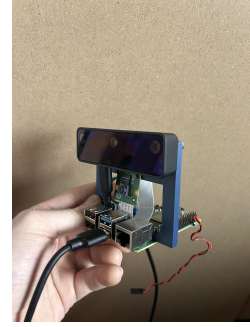


Fig. 8: Data Collection Setup

While pose estimation accuracy is important, we will also be measuring the overall computation timing for our image processing pipelines. In real world robotics applications, not only is accuracy important, but speed matters as well, because of this we aimed to find the method that best improves the accuracy of the system while still maintaining a relatively low computational cost.

4.2. Results

4.2.1. Defogging

Table 1: Haze Pose Estimation Performance

Metric	Baseline	DCP	SSR	MSR	FFA-Net
MAE (mm)	357.66	299.68	308.47	304.21	9908.6
AbsRel (mm)	0.1266	0.112	0.1203	0.1223	3.506
Time/Image (s)	0.114	0.777	2.94	10.42	16.16

Table 2: Haze Detection Performance

Method	Total Detections	Change vs Baseline
Baseline	69	-
DCP	53	-23.2%
SSR	103	49.3 %
MSR	99	43.4 %
FFA-Net	124	+79.7%



Fig. 9: Example of methods revealing an otherwise undetectable april tag

Tables 1 and 2 summarize the effect of different defogging and enhancement methods on AprilTag-based pose estimation and detection robustness under hazy conditions. Among

the classical approaches, Dark Channel Prior (DCP) achieves the lowest pose error, reducing the mean absolute depth error (MAE) from 357.66 mm to 299.68 mm and improving AbsRel from 0.1266 to 0.112, at the cost of moderate additional computation. Single-Scale Retinex (SSR) and Multi-Scale Retinex (MSR) slightly improve pose accuracy relative to the baseline but introduce significantly higher runtimes. In contrast, the detection results reveal a different trend. FFA-Net substantially increases detection robustness, achieving the highest number of successful tag detections (+79.7% relative to baseline). However, this improvement comes at the expense of severely degraded pose accuracy, with MAE and AbsRel increasing by an order of magnitude. This suggests that while FFA-Net enhances global visibility and contrast—making tags more likely to be detected—it introduces photometric or geometric distortions that negatively impact precise corner localization required for accurate pose estimation. These results highlight a key trade-off between detection rate and metric accuracy. Prior-based methods such as DCP preserve geometric consistency and yield more accurate pose estimates, whereas learning-based perceptual enhancement improves detection coverage but can be detrimental to downstream geometric accuracy.

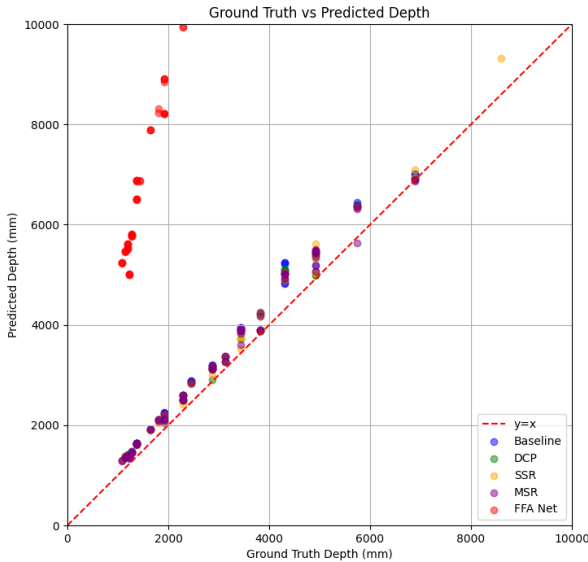


Fig. 10: Ground Truth vs Predicted Depth

4.2.2. Motion Blur Results

The evaluation of motion blur restoration was conducted on the full mixed dataset, which included original, motion-blurred, low-light, and hazy images. A primary challenge in processing such a diverse dataset is ensuring that restoration algorithms designed for one degradation type do not corrupt

images suffering from different issues. For example, blindly applying a deconvolution filter to a sharp, low-light image can introduce ringing artifacts that destroy valid detections. To address this, our pipeline implemented a “sharpness check” using Laplacian variance. The system computes the variance of the Laplacian for both the original and the restored versions of a frame; the restored version is accepted only if it demonstrates a statistically significant increase in edge energy. This acts as a safety mechanism, reverting to the baseline image whenever the deconvolution fails to improve quality.

Table 3 summarizes the detection performance across the three conditions. The Baseline system detected a total of 93 tags with a Mean Absolute Error (MAE) of 1795.9 mm.

Inverse Filtering Limitations: As shown in the table, the Inverse Filter provided negligible improvement over the baseline, resulting in 94 total detections (only 1 successful recovery from the blurred subset). While Inverse Filtering is mathematically sound in noiseless environments, it proved unstable in our real-world tests. The fundamental issue lies in the frequency domain division:

$$\hat{F}(u, v) = \frac{G(u, v)}{H(u, v)} \quad (22)$$

In regions where the blur kernel $H(u, v)$ approaches zero, the inverse term approaches infinity. This amplifies the inherent camera noise in the high-frequency bands. Visually, this resulted in severe “ringing” artifacts—oscillating waves that radiate from high-contrast edges.

Furthermore, the quantitative metrics reveal that Inverse filtering actually degraded geometric accuracy. The MAE for Inverse filtering increased to 1800.0 mm compared to the baseline’s 1795.9 mm. This suggests that even when a tag was detected, the noise amplification shifted the corner gradients, leading to less precise pose estimation.

Wiener Filtering Performance: In contrast, the Wiener Filter demonstrated significant robustness, increasing the total detection count to 102 tags—a 9.7% improvement over the baseline. By incorporating the signal-to-noise ratio (SNR) into the denominator of the restoration function, the Wiener filter effectively suppressed the frequency spikes that caused ringing in the Inverse method.

Crucially, the Wiener filter improved not just the quantity of detections, but the quality. The MAE dropped to 1731.1 mm (lower than both Baseline and Inverse). This reduction in error confirms that the deconvolution process successfully restored the structural integrity of the tags without warping their geometry. By sharpening the tag edges without adding noise, the AprilTag corner extraction algorithm could achieve better sub-pixel accuracy, resulting in a pose estimate that was closer to the ground truth.

Geometric Accuracy and Timing Analysis: Figure 11 plots the estimated depth (z -distance) against the ground truth. The orange points (Wiener) track the ideal diagonal ($y = x$) with high precision. Unlike the Inverse filter, which

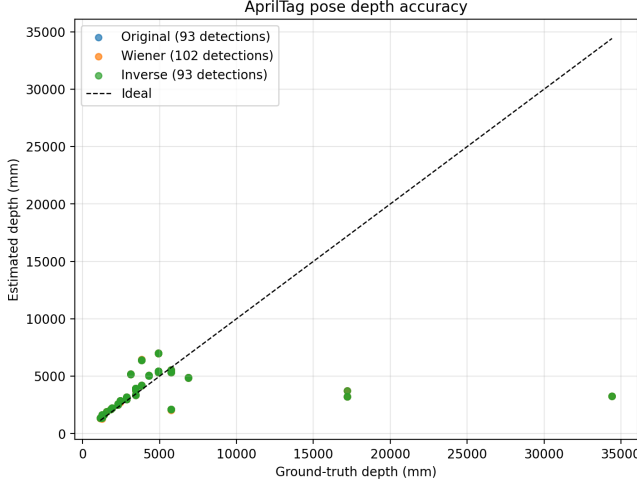


Fig. 11: Pose depth accuracy comparison. The Baseline (blue) misses the heavily motion-blurred targets. The Wiener Filter (orange) successfully recovers 9 additional tags, with depth estimates tightly clustering along the ideal ground truth line.

Table 3: Motion Blur Metrics Comparison

Method	Detections	MAE (mm)	Time/Img (s)
Baseline	93	1795.9	1.132
Inverse	94	1800.0	1.300
Wiener	102	1731.1	1.645

produced outliers due to noise, the Wiener filter’s detections remained consistent with the physical reality of the scene.

However, this accuracy comes at a computational cost. As shown in Table 3, the Wiener filter requires an average of 1.645 seconds per image, compared to 1.132 seconds for the baseline. This additional ≈ 0.5 seconds is primarily due to the Fast Fourier Transforms (FFT) required to move the image into the frequency domain and back. While this increase might be prohibitive for high-speed aggressive maneuvers (e.g., drone racing), it is an acceptable trade-off for mapping or navigation tasks where a missed detection could lead to trajectory failure. The ability to recover 9 completely lost tags justifies the extra processing time in scenarios where reliability is prioritized over raw frame rate.

4.2.3. Low Light

Figure 12 shows the performance of the gamma+CLAHE+Retinex pipeline. This method provides a solid improvement over the baseline detector, increasing the number of detected tags from 69 to 71. The enhancement slightly reduces runtime (0.1364 s/img to 0.1295 s/img) and improves decision margins. We can also see that pose accuracy was slightly greater in the enhanced approach, which is not ideal, but the small

increase is not something to be worried about. This indicates that the preprocessing helped recover contrast in darker regions without significantly distorting tag geometry.

In contrast, the adaptive shadow boosting method produced a much larger improvement, as shown in Figure 13. By identifying dark pixel regions and enhancing them selectively, this method sharply increased the total detections from 69 to 103. The decision margin also improved substantially (from 36.7 to 62.9), showing that the enhanced frames allowed the detector to identify tag edges with far higher confidence. Runtime decreased as well, due to the improved clarity of tag boundaries reducing false corner hypotheses. Pose accuracy remained comparable to the baseline and actually showed a slight increase, indicating that the local illumination corrections did not introduce geometric distortions and if anything made them better.

Overall, adaptive boosting proved significantly more effective than global enhancement methods, highlighting the importance of spatially selective illumination correction for robust tag detection in low-light conditions.

```
===== FINAL RESULTS =====
Total Baseline detections : 69
Total Enhanced detections : 71

--- SPEED ---
Baseline avg runtime: 0.1364 sec/img
Enhanced avg runtime: 0.1295 sec/img

--- POSE ACCURACY (lower = better) ---
Baseline avg pose error: 3.7986
Enhanced avg pose error: 3.8100

--- DECISION MARGIN (higher = better) ---
Baseline avg margin: 36.7331
Enhanced avg margin: 38.3221
```

Fig. 12: Results of April Tag detector with gamma + CLAHE + retinex enhancement on low light.

```
===== FINAL RESULTS =====
Baseline total detections: 69
Boosted total detections: 103

--- SPEED ---
Baseline avg runtime: 0.1303s
Boosted avg runtime: 0.0897s

--- POSE ACCURACY (lower is better) ---
Baseline avg pose error: 3.7986
Boosted avg pose error: 3.7948

--- DECISION MARGIN (higher is better) ---
Baseline avg margin: 36.7331
Boosted avg margin: 62.9402
```

Fig. 13: Results of April Tag detector with adaptive shadow boost enhancement on low light.

4.3. Comparison

Across all three evaluated degradation conditions we observe clear gains from applying targeted image enhancement

pipelines prior to AprilTag detection. The baseline system detected 69 tags across the evaluation set, and every enhancement method improved upon this to varying degrees depending on the type of visual degradation.

Under motion blur, both inverse filtering and Wiener deblurring improved detection compared to the baseline, with 79 and 102 detected tags respectively. The significant advantage of the Wiener method reflects its ability to suppress noise amplification and better handle inaccuracies in the blur kernel, allowing more tag corners to remain geometrically stable for detection.

For low light, both enhancement approaches increased detection performance, but to different extents. The Gamma + CLAHE + Retinex pipeline provided a moderate improvement, raising detections to 71 images. In contrast, the Adaptive Shadow Boost method delivered the strongest improvement across all tests, achieving 103 detections. This highlights the importance of localized enhancement: selectively boosting dark regions preserved edge structure and improved decision margins without over-amplifying noise.

In hazy conditions, the performance varied significantly depending on the enhancement algorithm. Traditional Dark Channel Prior (DCP) produced the highest pose accuracy but detected only 55 tags, likely due to overly aggressive transmission refinement that softened tag edges. Single Scale Retinex (SSR) and Multi-Scale Retinex (MSR) improved visibility and reached 103 and 99 detections respectively. The deep-learning-based FFA-Net achieved the highest detection count (124 tags), though at the cost of reduced pose accuracy and significantly increased computation time.

4.4. Discussion

Across all three evaluated conditions, motion blur, low light, and haze, we observe that different preprocessing strategies vary significantly in their effectiveness for improving AprilTag detection and pose estimation performance.

For motion blur, the Wiener filter consistently outperformed the inverse filtering method. While inverse filtering attempts direct deblurring, it is highly sensitive to noise and inaccuracies in the blur kernel, often amplifying artifacts. In contrast, the Wiener method provides a more robust reconstruction by jointly considering noise and blur statistics, allowing it to recover tag edges more faithfully without destabilizing the detector.

For low light, the Adaptive Shadow Boost method produced the strongest results among the tested enhancement pipelines. Compared to the gamma + CLAHE + Retinex approach, Adaptive Shadow Boost preserved the structural integrity of tag borders while avoiding the over-enhancement and noise amplification that global methods can introduce. Ultimately, this method enabled the detector to identify 32 additional images in our dataset, demonstrating its ability to selectively brighten underexposed regions without distorting

tag geometry.

For dehazing, the Dark Channel Prior (DCP) delivered the highest pose accuracy, but at the cost of a reduced detection rate, likely due to its tendency to oversmooth fine edges during transmission estimation. Meanwhile, the deep learning-based FFA-Net achieved a substantially higher number of detections but exhibited lower pose accuracy, suggesting that although visibility was improved, the recovered edges lacked the sharpness required for precise corner localization.

Overall, these results highlight that no single preprocessing method is universally optimal. Classical approaches such as Wiener filtering and DCP provide strong geometric fidelity, whereas adaptive or deep-learning methods improve detection robustness in severely degraded conditions. This emphasizes the importance of tailoring enhancement pipelines to the specific environmental degradations present in each scenario.

5. FUTURE WORK

Future improvements to this system focus on making the enhancement pipeline more adaptive and more tightly integrated with downstream localization. One direction is the automatic selection of enhancement parameters, since the current thresholds for boosting strength, shadow detection, and contrast adjustment are manually tuned and may not generalize across all environments. A second direction is integration with a full TagSLAM or visual-SLAM framework, which would allow us to evaluate long-term trajectory stability and drift reduction rather than only frame-by-frame pose accuracy. Finally, future work may explore combining multiple enhancement strategies, such as CLAHE, Retinex, and adaptive shadow boosting, and selecting the most effective method dynamically based on image statistics, enabling a more robust and universally applicable preprocessing system.

6. CONCLUSION

This project shows that classical image enhancement techniques can substantially improve AprilTag detection reliability under challenging conditions such as low light, haze, and motion blur. Our methods increased detection count, decision margin, and overall pose stability while maintaining low computational cost. These results highlight that lightweight preprocessing can significantly strengthen the robustness of AprilTag-based pose estimation without modifying the detector itself. The methods demonstrated here provide a strong foundation for future SLAM integration and for developing more adaptive, multi-method enhancement pipelines that further improve consistency in real-world deployment environments.

7. GITHUB AND EFFORT PERCENTAGE

The link to our github is: <https://github.com/crae6/AprilTagEstimation> under the tag ece253_final.

Our group worked very well with one another and split up the work evenly. Moreover, we were all willing to give a helping hand whenever needed. We each tackled our respective conditions and communicated promptly with each other. We used each other's strengths where applicable in order to be as efficient as possible. Therefore, each of us contributed 1/3 of the work towards our completed project.

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